

THE IMPACT OF A LARGE DEPRECIATION ON THE COST OF LIVING OF RICH AND POOR CONSUMERS*

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Using retailer scanner data, we investigate how a sharp and abrupt depreciation of the exchange rate affects consumers' cost of living. We find that the marginal cost of imported goods increased, whereas their retail markups decreased compared to domestic products. There was also a surge in the entry and exit of both foreign and local product varieties post-depreciation. Wealthier consumers, who spend more on imports, are hit harder by higher marginal costs but benefit from reduced markups and increased product diversity, unlike their less affluent counterparts.

1. INTRODUCTION

Developing countries often experience considerable terms of trade volatility and nominal exchange rate fluctuations. In response, a growing literature has tried to understand how such international shocks affect the average cost of living and how consumers across the income distribution are differently affected.

The lack of detailed consumer-level price and quantity data has, however, precluded the literature from studying certain mechanisms through which international shocks transmit into final consumer prices and the cost of living. First, although there is abundant evidence that distribution margins matter for explaining the disconnect between consumer and border prices (Burstein et al., 2003; Hellerstein, 2008; Berger et al., 2012), considerably less is known about how distribution margins change in response to international shocks. In particular, it is unclear whether changes in distribution margins dampen relative price adjustment following a currency depreciation and alter the distributional price effects of international price changes. Second, the literature on the distributional effects of international shocks has focused predominantly on quantifying how heterogeneity in expenditure shares across product categories leads to distributional cost-of-living effects (Fajgelbaum and Khandelwal, 2016; Cravino and Levchenko, 2017). However, category-level price effects originate from changes in the prices of continuing varieties and from changes in the underlying set of available varieties. If consumers have heterogeneous preferences over foreign and local varieties, relative price changes and changes in product availability may affect rich and poor consumers differently.

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This article studies how consumer prices, costs, retail margins, and product availability change following a large nominal exchange rate shock and how these adjustments induce distributional cost-of-living effects. We exploit the depreciation of the Kazakh Tenge (KZT) in August 2015 following the switch from a fixed to a floating exchange rate regime that occurred against the backdrop of falling international commodity prices. This episode provides us with an intriguing setting for three reasons. First, the depreciation was sudden, allowing us to demarcate a clear pre- and post-depreciation period. Second, the depreciation was substantial, trumping most concurrent shocks, and came after a period of foreign exchange stability due to the fixed exchange rate that was in place before the depreciation.¹ Third, we study the distributional effects of foreign exchange shocks on consumer prices in the context of an emerging economy. Data availability has confined most of the literature to either study advanced economic settings in a very detailed manner (Borusyak and Jaravel, 2021) or to focus on heterogeneity across product categories in developing economies (Cravino and Levchenko, 2017).² As the average quality of imports and exports differs markedly between developed and developing countries (Schott, 2004), the exposure of rich and poor consumers to international shocks likely differs between high- and low-income countries as well.

We draw on highly detailed scanner data from a supermarket chain, Metro, at the product and the transaction level. The product-level data provide us with price, quantity, and cost data for both local and foreign products within highly detailed product categories. Observing both price and cost at the product level enables us to examine how retail margins behave in response to a currency depreciation without resorting to strong structural assumptions on demand, supply, or market structure. We use the accompanying transaction-level data to subdivide consumers into different income groups by leveraging quality Engel curves, that is, the fact that richer consumers tend to purchase products with higher unit prices. This allows us to study how the cost of living evolves differently for rich and poor consumers.

We focus on purchases of food and nonalcoholic beverages and start by providing three pieces of reduced-form evidence. First, within detailed product categories, rich consumers spend on average more on foreign varieties relative to poor consumers. This heterogeneity in expenditure shares implies that, within product categories, rich consumers were more exposed to changes in the relative price of foreign varieties.

Second, consumer prices rose on average by 20% following the depreciation. However, within detailed product categories, consumer prices of foreign varieties increased only by 3–4% more relative to local varieties. Whereas marginal costs increased by 7–8% more for foreign varieties than for local varieties, retail margins on foreign varieties fell by 3–4%. Using additional data, we show that it is unlikely that the markup adjustments were a response to competitive pressure in the retail sector as prices at other stores changed comparably. Rather, we argue that the markup adjustments are consistent with an optimal response by a multiproduct retailer which, consequently, altered the transmission of the shock into consumer prices.

Third, whereas relative consumer price adjustment was fairly weak, we document considerable changes on the extensive margin. Following the depreciation, we identify a break in the entry and exit patterns for foreign and local varieties as the exit rate of foreign varieties and the entry rate of local varieties increased. This underscores that, within product detailed product categories, distributional effects of currency depreciations could also arise through changes at the extensive margin if, for instance, rich and poor consumers vary in their perception of the substitutability of different varieties.

To understand whether changes in relative prices and product availability actually led to distributional cost-of-living effects, we model consumer preferences according to a nested

¹ After one, three, and six months, the currency had lost 36.9%, 55.9%, and 78.5% of its value to the U.S. Dollar (USD). Auer et al. (2021) study a similar setting with the sudden appreciation of the Swiss Franc in 2015.

² One exception is Atkin et al. (2018) which study the aggregate and distributional welfare effects of retail FDI in Mexico using similarly detailed data.

mixed-Constant-Elasticity-of Substitution (CES) demand system. The mixed-CES specification is well-suited to capture the three pieces of reduced-form evidence. First, by allowing parameters and budget shares to vary across income groups, it provides a nonparametric way to account for preference heterogeneity between rich and poor consumers (Atkin et al., 2018; Jaravel, 2019; Argente and Lee, 2021). Second, mixed-CES systems are consistent with relative shifts in markups following relative changes in marginal costs (Redding and Weinstein, 2020). Finally, the CES family of demand systems is the workhorse framework to quantify the effect of product entry and exit (Feenstra, 1994; Broda and Weinstein, 2006).

This structure allows us to decompose changes in the cost of living into three channels: (i) the price channel, consisting of a markup and a cost effect, (ii) the substitution channel, and (iii) the product variety channel. The decomposition highlights the three ways in which consumers can be differently affected following an international shock. First, if consumers differ in their pre-shock expenditure shares on varieties whose prices adjust differently following the depreciation, their cost of living adjusts differently. Second, if consumers differ in their responsiveness to price changes, it will change their exposure *ex post*.³ Finally, the variety channel captures that if consumers differ in how substitutable they perceive varieties or if they switch differently from existing varieties toward entering varieties, they will also be differently affected. To quantify these different channels, we estimate the elasticities of substitution by making use of the variety-level price and quantity data. We document substantial differences between rich and poor consumers. Across all specifications, we estimate that the price elasticity of demand is, on average, twice as high for poor consumers compared to rich consumers.

We find that, one year after the depreciation, the cost of living for the same basket of food and nonalcoholic beverages increased by 20%. Whereas the cost of living went up by 24% for poor consumers, it only went up by 16% for rich consumers. First, consistent with the reduced-form evidence, rich consumers experienced a larger increase in the marginal cost of food and beverages, but this was offset by a reduction in the average retail markup. Second, because poor consumers have higher elasticities of substitution, they reallocated expenditure more toward varieties that experienced a smaller price increase following the depreciation which dampened the increase in their cost of living somewhat. Finally, whereas rich and poor consumers spent similarly on entering relative to exiting varieties, the lower elasticities of substitution for rich consumers imply that they experienced a lower increase in the cost of living. Altogether, because the price and substitution channels offset each other and render the intensive margin distributionally neutral, we find that the extensive margin drove the distributional effects.

This article contributes to two streams of literature in several ways. First, we contribute to the literature on the exchange pass-through. The literature on exchange rate pass-through into import prices is vast (Gopinath and Rigobon, 2008; Gopinath et al., 2010; Berman et al., 2012; Amiti et al., 2014, 2019) but connecting currency fluctuations to consumer prices has been elusive. Most papers either rely on price indices (e.g., Burstein et al., 2005; Goldberg and Campa, 2010) or link import and consumer prices at the product category level (Berger et al., 2012; Auer et al., 2021). Using data on consumer prices, retail markups, and marginal retail costs for foreign and local products at the variety level, we contribute to this literature in two ways. First, we provide novel evidence on how changes in retail markups offset the relative cost increase for foreign varieties. Second, recent work by Nakamura and Steinsson (2012), Cavallo et al. (2014), Goetz and Rodnyansky (2023), and Crowley et al. (2024) shows how firms use product introductions and replacements as a source of price flexibility in response to currency fluctuations. We document considerable changes in product availability and show that accounting for them is crucial to understanding how the cost of living changes following exchange rate fluctuations.

Second, we contribute to the literature on distributional cost-of-living effects in response to international shocks, such as trade liberalization (Porto, 2006; Faber, 2014; Fajgelbaum and

³ This corresponds to the unequal expenditure switching channel highlighted in Auer et al. (2023).

Khandelwal, 2016) and currency devaluations (Cravino and Levchenko, 2017; Auer et al., 2023). Cravino and Levchenko (2017) distinguish between distributional effects stemming from differential exposure across product categories and from heterogeneity in the adjustment of prices within product categories. This article contributes to this second strand in the literature in two ways. First, this literature has mostly focused on the role of heterogeneity in expenditure shares across varieties in generating distributional effects across rich and poor consumers. Like Auer et al. (2023), we also consider the implications of differences in the price sensitivity across rich and poor consumers for evaluating the impact of international shocks on consumer welfare. Whereas Auer et al. (2023) focus on differences in substitution across consumer groups, this article emphasizes that if consumers differ in their elasticities of substitution, they will also be differently affected through changes on the extensive margin. Second, data limitations have mostly precluded the literature from studying how international shocks transmit into final consumer prices and how they might affect them in heterogeneous ways. This article relies on data on consumer prices, retail markups, and costs to document how international cost shocks are partially offset through adjustments in the retail markups and the extensive margin of product varieties.

The rest of this article is organized as follows: Section 2 introduces the data. Section 3 documents the three pieces of reduced-form evidence that motivate studying the distributional cost-of-living effects following the depreciation. In Section 4, we develop the framework, and Section 5 presents the changes in the cost of living. Finally, Section 6 concludes.

2. DATA

We use matched store- and consumer-level scanner data from a large retailer, Metro, in Kazakhstan. Metro entered the Kazakh market in 2009 and currently operates eight stores across the country. The data cover two stores, one in Almaty, the economic capital, and one in Nur-Sultan, the official and administrative capital. The data are collected through daily price scans between September 2014 and November 2017. Metro's product assortment covers food and nonalcoholic beverages, tobacco and alcoholic Beverages, household tools (cleaning, cooking tools, etc.), and clothing. We restrict attention to frequently shopping households at the consumer level and focus on food and nonalcoholic beverages. We provide more details below.

2.1. Transaction-Level Data. A transaction contains a unique customer ID, the product that was bought, the total expenditure, the number of units bought, the store at which the product was bought, and the time stamp. To shop at the stores, store policy dictates that customers are required to have a loyalty card. We rely on consumer-level transaction data to subdivide consumers into different income groups based on quality Engel curves.

Income definition Ideally, we subdivided consumers based on their reported income. As the retailer did not collect this information, we do not observe consumer-level income. There is, however, substantial evidence that richer consumers tend to consume the higher-priced varieties within product categories. In other words, consumers adhere to so-called quality Engel curves. Deaton (1988) shows this regularity for food purchases in Ivory Coast and Bils and Klenow (2001) confirm this for U.S. households. This pattern also holds in more recent data as shown by Handbury (2021), Argente and Lee (2021), and Faber and Fally (2022) for the United States and by Cravino and Levchenko (2017) for Mexico.

We build on this evidence and subdivide consumers into three income groups based on the observed consumption patterns. Doing so, we proceed in four steps. First, we select the transaction data before the depreciation to avoid introducing a bias in the income group

classification.⁴ Second, we use package size information from the article name to express consumer prices in equivalent units (e.g., KZT per ml or kg). Table A.1 shows the share in terms of total expenditure and the number of varieties of each unit across product categories. For example, varieties in the subcategories “soft drinks” and “water” are almost solely expressed in milliliters, whereas subcategories “fish” and “meat” are in grams.⁵ Third, within each product group, we rank varieties according to their median pre-depreciation unit price and categorize varieties into four types: (i) very cheap, (ii) cheap, (iii) expensive, and (iv) very expensive varieties based on the product group-specific quartiles:

$$(1) \quad f(p_{i,g}^{\text{med}}, P_g) = \begin{cases} 1, & \text{if } \mathcal{P}(P_g \leq p_{i,g}^{\text{med}}) \leq 0.25, \\ 2, & \text{if } 0.25 < \mathcal{P}(P_g \leq p_{i,g}^{\text{med}}) \leq 0.5, \\ 3, & \text{if } 0.5 < \mathcal{P}(P_g \leq p_{i,g}^{\text{med}}) \leq 0.75, \\ 4, & \text{if } \mathcal{P}(P_g \leq p_{i,g}^{\text{med}}) > 0.75, \end{cases}$$

where $p_{i,g}^{\text{med}}$ is the pre-depreciation median unit price of a variety i in product group g and P_g is the random variable representing pre-depreciation unit prices in product group g . Finally, we compute for each consumer an index that is the weighted average of how expensive her consumption basket is as follows:

$$\text{Index}_j = \frac{\sum_g \sum_i \sum_{t \leq 2015Q2} f(p_{i,g}^{\text{med}}, P_g) \cdot p_{i,g,t} \cdot q_{j,i,g,t}}{\sum_g \sum_i \sum_{t \leq 2015Q2} p_{i,g,t} \cdot q_{j,i,g,t}}.$$

We define poor consumers as consumers who have an index value in the first quintile; rich consumers have an index value in the fifth quintile. Figure 1 shows the distribution of the index and indicates the different income groups in separate colors.⁶ Below, we check the robustness of the distributional effects by considering alternative cutoffs.

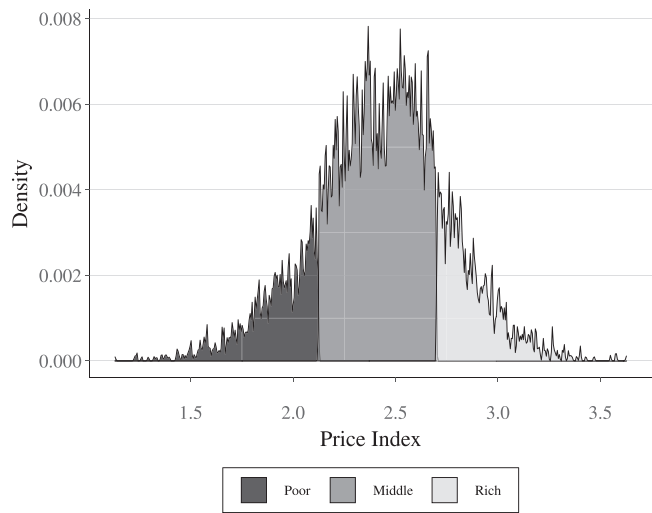
An alternative way to infer income is to classify consumers based on expenditure per capita (Faber and Fally, 2022). Although we observe total expenditure, we do not have information about the size of the household. Hence, it is unclear whether higher expenditure reflects higher income or larger household size. For this reason, we prefer quality Engel curves. Nonetheless, the main results of the article are robust to using total expenditures as the income classification method.

Households The consumer pool of Metro comprises both households and small business owners, such as restaurants and small shops. To isolate households from small business owners, we discard expenditures that are unlikely to be made by households. Table A.2 shows the average expenditure per month and the corresponding average expenditure per week in local currency and in USD. Given that the average monthly wage in Kazakhstan was 126,021 KZT

⁴ As the depreciation increased the price of foreign products, including the prices of products after the depreciation could bias the income group classification. One implication of this choice is that we cannot compute this index for consumers who start buying after the depreciation. This restriction is not problematic for our purposes as we are interested in the evolution of the cost of living relative to the cost of living before the depreciation and we would need to exclude these consumers anyhow.

⁵ There are four different levels of categorization in the data set: (i) Categories (e.g., food), (ii) Subcategories (e.g., fruit), (iii) Product groups (e.g., stonefruit), and (iv) Products (e.g., peach) (see *infra*). We have chosen to conduct the exercise at the product group level to make sure that we have a sufficient number of articles to compute the distributions.

⁶ In the construction of these income groups, we pool across consumers shopping in the different stores. Figure A.1 shows that the distribution of this index is very similar across stores and therefore we can safely aggregate across stores.



NOTES: This figure displays the distribution of the index that captures the weighted average of how expensive consumers’ consumption basket is. Poor consumers are defined as consumers that have an index in the lowest quintile of the distribution; rich consumers have an index in the highest quintile of the distribution. The index is constructed by only including expenditure before the depreciation and is pooled across stores.

FIGURE 1
DISTRIBUTION OF INDEX (QUINTILES: 20–80% SPLIT)

(or 568 USD) in 2015, we exclude consumers who rank above the 99th percentile of the distribution of the average monthly expenditures.⁷ Also, because we study the evolution of the cost of living during the depreciation episode, we remove all consumers who did not shop at the retailer before the depreciation.⁸

Frequently shopping households To quantify the effect of the depreciation on the cost of living of different consumers, we need to track the consumption patterns of the same consumers over time. Therefore, we only focus on consumers who frequently shop at the store. To be part of the frequent sample, we require consumers to shop at the retailer for at least eight months out of the 11 months before the depreciation and at least nine months in the 12 months directly after the depreciation. Tables A.4 and A.5 compare the frequent and full samples on a set of observable characteristics. The frequent sample contains 5,040 consumers, who jointly account for 27% of total expenditures. Table A.5 shows that the consumers in the frequent and complete sample are almost identical in terms of price and composition of the consumption basket. The results are even more pronounced when we include consumers of the infrequent sample as well.

2.2. Product-Level Data. We have access to rich product-level data that cover the full universe of products sold by the retailer. We observe the quantity and the price for each purchase made by the customers on a given day and the inventory value and quantity for each purchased variety at each point in time.⁹

⁷ Calculations based on data from the International Labor Organization (ILO).
⁸ In this way, we remove about one-third of the consumers (Table A.3). This group, however, only accounts for 23% in total sales and for 14% of the transactions.
⁹ Many papers identify a product by recording data at the barcode or UPC level (Hottman et al., 2016; Jaravel, 2019). Like in Anderson et al. (2015), we identify products at the stock-keeping unit which is at least as disaggregated as the UPC or EAN level as, in practice, the same UPC may be associated with more than one SKU.

Variable construction Observing the inventory value and quantity for each purchased variety, is crucial to our analysis for two reasons. First, the inventory information enables us to compute variety-level replacement or marginal costs. The retailer applies a first-in-first-out (FIFO) inventory principle to register sales and administer the inventory. We combine this knowledge with the inventory data and back out the cost that is attributed to the latest inventory restocking.¹⁰ This inventory cost includes the most recent wholesale price paid and the variable distribution costs necessary to put products on the store's shelves. We will refer to this cost measure as the marginal costs as the costs associated with buying and selling products are the most important part of retailers' cost structure. Like Gopinath et al. (2011), Eichenbaum et al. (2011), and Goetz and Rodnyansky (2023), we use this cost measure and define retail markups as the ratio of prices and replacement costs. Second, the inventory data provide direct information about which product varieties are available to consumers and thus which products enter and exit the choice set of consumers at the store.

Food and nonalcoholic beverages The analysis focuses on purchases of food and nonalcoholic beverages. To maximize the external validity of the results, we concentrate solely on product categories that exhibit a comparable price evolution to the corresponding Consumer Price Index (CPI) component. To this end, Figure A.2 contrasts the aggregate price evolution of the categories food and nonalcoholic beverages, tobacco and alcoholic beverages, household tools, and clothing with the price evolution of their corresponding CPI component.¹¹ Looking at Figure A.2, only food and nonalcoholic beverages show a price evolution that is similar before and after the depreciation. The other categories either miss the mark before, after, or before and after the depreciation. While the sample coverage inevitably reduces, column 3 of Table A.6 shows that food and nonalcoholic beverages are the most important source of revenue for the retailer representing 61% of its revenue. Moreover, Table A.7 indicates that food and nonalcoholic beverages account for 34% of the Kazakh CPI basket, making it the most important category in the overall CPI construction.

Foreign and local varieties We match the retailer's proprietary product identification number with the variety's EAN code provided by the retailer.¹² We follow Bems and Giovanni (2016) and subdivide varieties into foreign and local based on the EAN code: if the variety's EAN code starts with "487," Kazakhstan's country code, the variety is labeled as "local"; for any other code, the variety is labeled as "foreign." Table 1 shows the share of varieties that we can identify as either foreign or local and the foreign share that we obtain for different subcategories. In terms of expenditure, we classify around 80–90% of varieties as being local or foreign, apart from meat, vegetables, and fruits.¹³ Outside of the estimation of the event studies in Section 3, we include all varieties even if we cannot determine their origin.¹⁴

One potential problem with this approach is that foreign manufacturers might relabel their products or change the barcode of the product when they sell to a different market. One way to check the performance of the barcode-based classification method is to evaluate its accuracy in classifying varieties that are differentiated by their origin (Bems and Giovanni, 2016). To this end, we retrieved the origin country of wines from their barcode descriptions and found that the barcode classification method classified 97% of expenditure on foreign wines as foreign and 99% of expenditure on local wines as local.

¹⁰ Knowing the inventory principle is necessary to map the data on inventory costs to replacement costs or marginal costs (Peltzman, 2000).

¹¹ We compute a category-level price index from variety-level prices by computing a Törnqvist price index.

¹² In the EAN classification system, each barcode has 13 digits. The first three digits identify the country of registration of the manufacturer, the next five digits indicate the manufacturer, and the final five digits reflect the product.

¹³ These product categories are notoriously hard to classify and other papers usually discard these categories altogether (Cravino and Levchenko, 2017; Auer et al., 2021).

¹⁴ To conduct the event studies, we need to allocate a variety to the treated, that is, foreign varieties, or the control group, that is, local varieties. Therefore, we have to discard varieties whose origin we cannot determine.

TABLE 1
PRODUCT SAMPLE OVERVIEW

Subcategory	Sales Share	Variety $\times$ Store Units	Observations	Foreign Share	Classification Quality
Bakery/Cereal	0.05	3115	46,650	0.87	0.55
Candy	0.08	4395	60,683	0.89	0.68
Coffee/Tea	0.06	1208	25,120	0.97	0.82
Dairy	0.17	3292	57,447	0.82	0.55
Dry food	0.07	1933	34,406	0.88	0.49
Fish	0.05	1641	24,206	0.80	0.67
Fruit	0.04	1125	11,249	0.36	0.81
Meat	0.20	2384	27,009	0.05	0.39
Ready-made	0.01	541	8089	0.97	0.55
Savory	0.01	644	11,669	0.99	0.94
Seasoning	0.09	2347	40,407	0.88	0.54
Soft drinks	0.06	1606	30,106	0.99	0.73
Vegetables	0.07	1808	23,366	0.56	0.91
Water	0.03	231	5855	1.00	0.48

NOTE: This table provides an overview of different subcategories in food and nonalcoholic beverages. The column “Sales Share” indicates the share of each subcategory in total sales of food and nonalcoholic beverages. The column “Variety $\times$ Store” indicates the number of unique variety $\times$ store combinations in the data set. The column “Observations” indicates the number of months in which there were registered sales for a variety $\times$ store combination. The column “Foreign Share” shows the share of foreign products in total sales for that subcategory. Finally, the column “Classification Quality” indicates the percentage of sales in each subcategory we can classify as either foreign or local. All statistics are computed by pooling across time and stores.

Frequency To focus on the short- to middle-run effects of the depreciation, we aggregate the data to a monthly frequency by computing average consumer prices and costs and total sales and quantities within each month. Given the monthly frequency and that we focus on the consumption of food and nonalcoholic beverages by households, it seems reasonable to abstract from very short-run anticipatory effects before the depreciation. For the decomposition results, we will present results at the quarterly level but the decomposition results are very similar when we use monthly data instead.

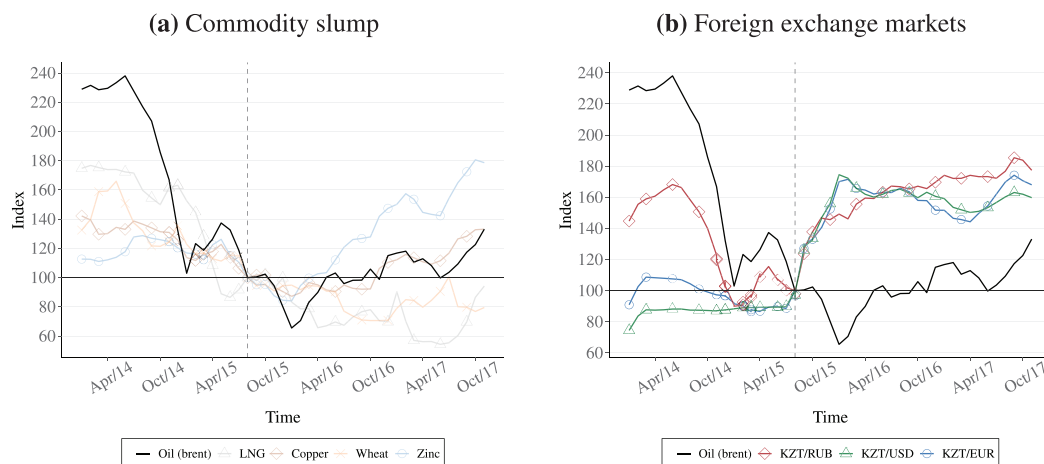
Representativeness of the store To support the external validity of our results, we provide an extensive analysis of the entire Kazakh retail sector in Appendix A.2. To this end, we combine scanner data from AC Nielsen, containing information about the 40 highest selling barcodes in each category, and data from the Kazakh National Bank (KNB).¹⁵ These data allow us to address two concerns regarding our approach to focus on one particular store. First, we show that prices for the same products at small and large stores, which together make up 85% of the retail sector, did not respond differently after the shock.¹⁶ Second, when studying the distributional consequences of depreciation, we need to make sure that we capture both rich and poor consumers at the store. We show that whereas small stores are cheaper and stock cheaper products, the price differences between varieties within our retailer are three times larger. This suggests that sorting of consumers across varieties within a store is likely more important than across stores.

3. REDUCED-FORM EVIDENCE

In this section, we discuss the depreciation of the KZT in August 2015 and its effect on consumer prices. We also provide three pieces of reduced-form evidence that motivate the estimation of changes in the cost of living for rich and poor consumers separately. First, rich consumers spend on average more on foreign varieties compared to poor consumers. Second,

¹⁵ The AC Nielsen data set contains barcode descriptions but does not contain EAN codes. Hence, we cannot match the AC Nielsen data to our data.

¹⁶ Metro has an overall market share of 10%.



NOTES: Panel (a) shows the evolution of the price for Brent crude oil, liquified natural gas (LNG), copper and zinc ore, and the global price for wheat. All series are normalized to their August 2015 level and were obtained from the St. Louis Federal Reserve database (FRED Database). Table A.12 shows that these commodities collectively made up around 80% of exports in 2015. In Panel (b), we repeat the series for Brent crude oil and show the evolutions of the Kazakh Tenge (KZT) versus the U.S. Dollar (USD), the euro (EUR), and the Russian Ruble (RUB). The foreign exchange series are taken from the IMF Financial database.

FIGURE 2

DEPRECIATION OF 2015

consumer prices of foreign varieties rose little relative to consumer prices of local varieties because changes in retail markups compensated for changes in costs. Finally, there was a substantial adjustment on the extensive margin with the increased entry of local varieties and exit of foreign varieties.

3.1. The Depreciation and Consumer Price Inflation. Kazakhstan is an emerging economy that primarily exports commodities and takes world prices as given (see Table A.12).¹⁷ Because the economy strongly relies on commodity exports, the KNB implemented a fixed exchange rate regime and pegged the Tenge to the euro and the USD before August 2015. Following the collapse of global commodity prices and the appreciation of the Tenge relative to the Russian Ruble (RUB) in 2015, the KNB switched to a floating exchange rate regime on August 20, 2015. As a result, the KZT sharply depreciated by 40% and 80% relative to a set of major currencies within six months as shown in Figure 2(b).

This episode provides an intriguing setting to study the transmission of exchange rate shocks into consumer prices and the cost of living for three reasons. First, the depreciation was sudden and persistent. Figure 2(b) shows that the Tenge depreciated immediately after the policy shift and stabilized at a much lower value after six months. This allows us to identify a clear pre- and post-event window. Second, the monetary policy shift was a response to external events. To support its fixed exchange rate policy, the KNB relied heavily on foreign currency inflows through the country's exports of commodities that tend to be denominated in foreign currency.¹⁸ However, Figure 2(a) shows that world prices of the economy's five most exported commodities started falling in late 2014. This made it increasingly difficult for the

¹⁷ Oil is Kazakhstan's largest exported commodity but its production only accounts for 1% or 2% of global oil production.

¹⁸ Gopinath (2015) and Boz et al. (2022) provide evidence that world prices of most commodities tend to be denominated in USD.

KNB to keep the nominal exchange rate at parity.¹⁹ Eventually, the KNB was forced to let go of the pegged exchange rate and decided to float the exchange rate, resulting in substantial exchange rate variation shown in Figure 2(b). Finally, the disaggregated nature of our data set makes it possible to plausibly demarcate the depreciation from other concurrent, and potentially confounding, events. We execute the analysis using granular scanner data at the individual variety and consumer level for which we obtain data on the origin of product varieties. As foreign varieties are exogenously more exposed to the depreciation, we can examine whether consumer prices, costs, retail markups, and product availability changed differently for foreign versus local varieties.

To stress the substantial effect of the depreciation on consumer prices, we estimate the following hedonic price regression:

$$(2) \quad \ln(p_{i,st}) = \theta_{p(i)o(i)} + \theta_{p(i),s} + \lambda_t + \varepsilon_{i,st},$$

where $\ln(p_{i,st})$ is the natural logarithm of the consumer price of a variety i in a store s at time t . We include category–origin fixed effects, $\theta_{p(i)o(i)}$, to filter out persistent price differences between products and origins.²⁰ We also add category–store fixed effects, $\theta_{p(i),s}$, that control for persistent differences across locations within product categories. Finally, we include λ_t which are time fixed effects that are normalized relative to August 2015. Figure 3 plots these time effects and underscores the inflationary effect of the depreciation. Consumer prices rose by 20% after six months and reached a new mid-to-long equilibrium level that was 25% higher after roughly one year. Because prices converged after 12 to 15 months, we focus on the first five quarters after the depreciation when we quantify the cost-of-living effects below.²¹

3.2. Heterogeneity in Foreign Expenditure Shares. A first step to understanding whether the depreciation potentially had distributional effects is to see whether rich and poor consumers spent differently on foreign products before the depreciation. We find that richer consumers indeed had on average a significantly higher expenditure share on foreign varieties both across and within detailed product categories.

Figure 4 shows the conditional distribution of the total expenditure share spent on foreign varieties for rich and poor consumers separately. There is a substantial shift in the distribution toward more spending on foreign varieties by rich consumers. Moreover, parametric and nonparametric tests reject the null hypothesis that the conditional distributions are stochastically equivalent (see Tables A.14 and A.15, respectively).²² Reassuringly, Figure A.7 shows that the disparity in the conditional distributions also monotonically increases when we move from looser toward stricter definitions of rich and poor consumers.

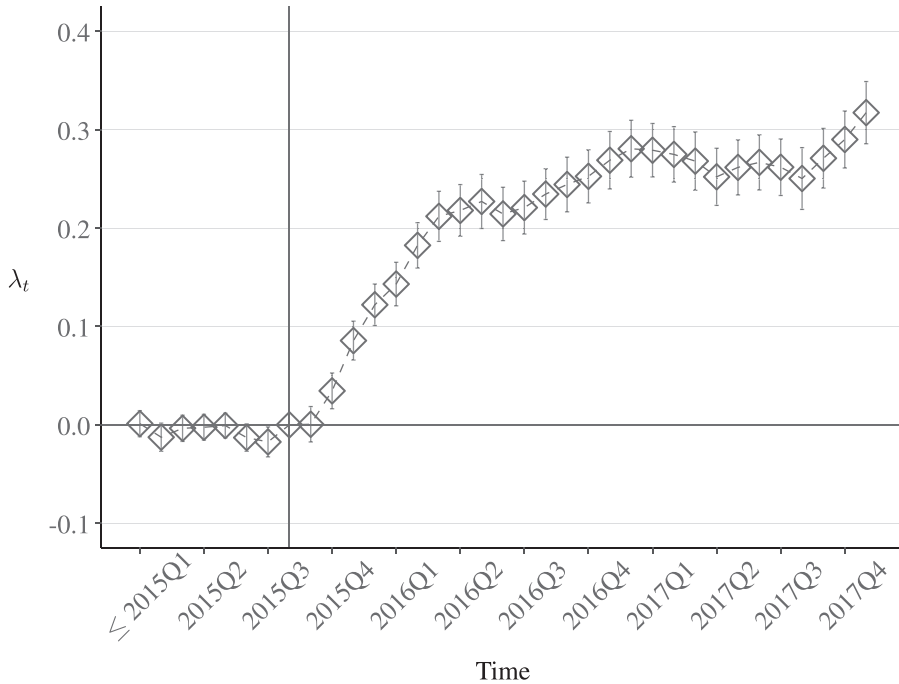
Figure A.9 illustrates that the same pattern also persists within the detailed product categories. Figure A.9 displays the conditional distributions of the share spent on foreign varieties across product categories for rich, poor, and in-between consumers. The distribution of category level of the foreign expenditure share is shifted upward for rich consumers and Tables A.14 and A.15 confirm that this pattern is statistically significant for all the different income

¹⁹ The drawdown of foreign currency reserves is confirmed in reports on the evolution of the KNB's balance sheet. In response, the National Bank increasingly borrowed foreign currency from other central banks to defend the fixed exchange rate as shown in Figure A.6.

²⁰ The category refers to the finest level of aggregation which is the product level. We interact category fixed effects with product origin fixed effects as Table A.9 shows that there are large price differences between foreign and local varieties.

²¹ Table A.13 shows that the Ruble, the euro, and the USD are the three main currencies of invoicing used by the retailer. Therefore, we considered whether there is interesting heterogeneity in the level of pass-through across currencies of invoicing. When we reestimate Equation (2) for different currencies of invoicing, we find that there is some heterogeneity across currencies in the transition toward the medium- to long-run pass-through level (see Figure A.10). Because the medium- to long-run pass-through level is very similar, we do not further explore this dimension.

²² We consider a parametric Kolmogorov–Smirnov and a nonparametric Paired-Rank-Sum-Wilcoxon test.



NOTES: This figure shows the evolution of consumer price following the depreciation. More specifically, we plot the coefficients λ_t which are obtained from estimating Equation (2). Note that we have bunched the effects prior to January 2015 into the effect for January 2015 and the effects after September 2016 into the effect of September 2016. Whiskers are 95% confidence intervals around the point estimates computed from standard errors which are clustered at the product-store level.

FIGURE 3

CONSUMER PRICES RESPONSE

group definitions. Like before, the pattern monotonically strengthens when we move from a loose to a strict income classification.

Classifying consumers based on total expenditure does not overturn this pattern. Figure A.8 illustrates that the conditional distribution of foreign shares for rich consumers still stochastically dominates the one for poor consumers when we subdivide consumers according to total expenditures.²³

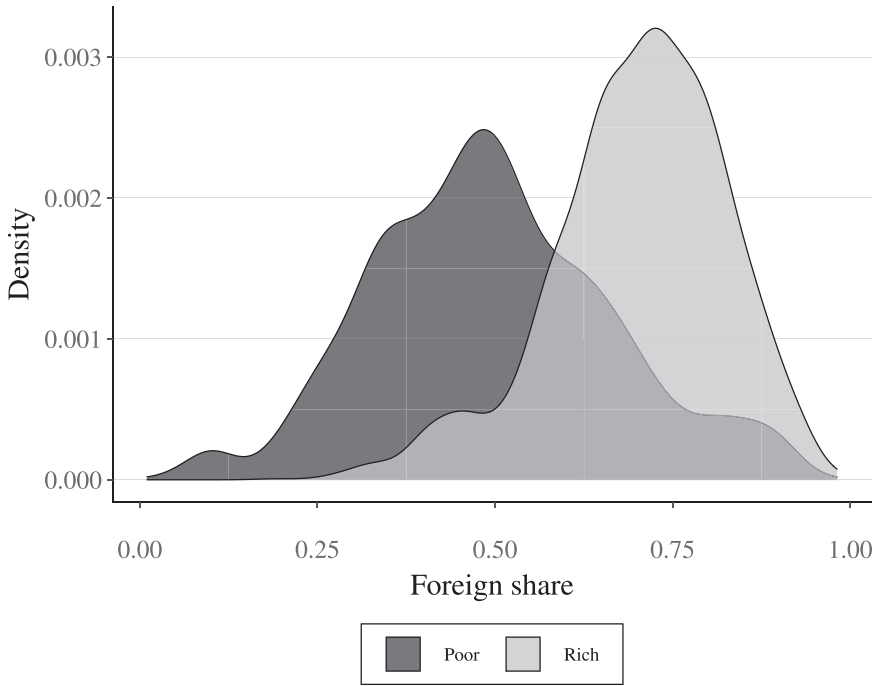
3.3. Relative Consumer Price Stability. Next, we estimate whether and by how much consumer prices of foreign varieties increased relative to consumer prices for local varieties. We also estimate how cost and retail markup for foreign varieties adjusted relative to local alternatives. To this end, we augment Equation (2) and estimate the following difference-in-difference specification:

$$(3) \quad y_{i,st} = \sum_t \beta_{q(t)} (\mathbb{1}(o(i) = \text{Foreign}) \times \mathbb{1}(q(t) \neq 2015Q3)) + \theta_{p(i)o(i)} + \lambda_{p(i),t} + \varepsilon_{i,st},$$

where $y_{i,st}$ is either the log price, log cost, or log retail markup of variety i in store s at time t , $\mathbb{1}(o(i) = \text{Foreign})$ is one when the origin of variety i is foreign and zero otherwise, and $\mathbb{1}(q(t) \neq 2015Q3)$ is one for all quarters apart from 2015Q3.²⁴ In this way, we estimate a

²³ Using this classification method, the Kolmogorov–Smirnov test also rejects the null hypothesis that the distributions are the same for rich and poor consumers.

²⁴ To include as many observations as possible, we consider observations at the store level as well. Below, we check the robustness of the results when we also consider fixed effects at the store level.



NOTES: This figure displays the distribution of the expenditure share on foreign varieties across rich and poor consumers separately. Income classification was executed using the expensiveness index. We include food and nonalcoholic beverages and the frequent sample of consumers in the construction of these graphs.

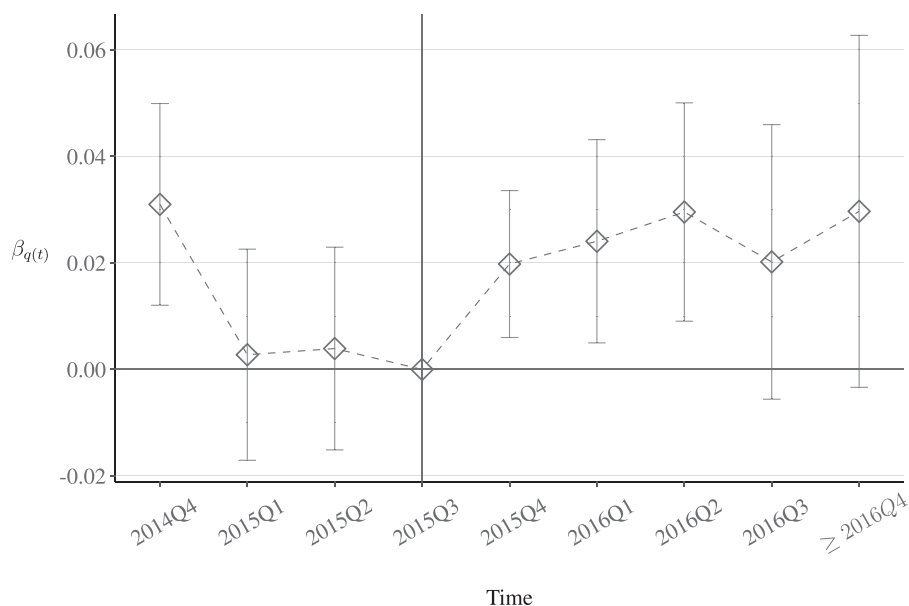
FIGURE 4
FOREIGN SHARE—PER INCOME GROUP (20–80% SPLIT)

pre-treatment effect for each quarter before 2015Q3 and a post-treatment effect for each quarter from 2015Q4 until 2016Q4. After 2016Q4, we collect the effects into one estimate.²⁵ We include two other sets of fixed effects. We keep the category–origin fixed effects $\theta_{p(i)o(i)}$ that control for persistent differences between foreign and local varieties at the category level. We substitute the time fixed effects for more detailed category–time fixed effects $\lambda_{p(i)t}$ that control for common changes at the category level for foreign and local varieties over time. To capture adjustment for continuing varieties, we only include continuing products and weigh their importance using pre-depreciation expenditure shares.

The coefficients of interest are the time-varying treatment effects, $\beta_{q(t)}$. Through foreign intermediate input sourcing or changes in manufacturing markups, local varieties were very likely affected by the depreciation as well. For this reason, the time-varying treatment effects only measure the differential adjustment of foreign varieties relative to local varieties and not the full effect of the depreciation on the prices of foreign varieties. Given that we are interested in how consumer prices of foreign varieties adjusted relative to local varieties, this difference-in-difference estimator is still suitable.

Consumer prices Figure 5 shows the results when we estimate Equation (3) for final consumer prices. We note that prices of foreign varieties were on a downward trend before the depreciation as one of the pre-depreciation treatment effects is statistically significant and positive. We believe this does not compromise the interpretation of the results for two reasons. First, the post-depreciation effects are not a continuation of the pre-depreciation trend as the depreciation induced a break from the pre-depreciation trend. The downward trend

²⁵ The results are very similar when we weight observations using sales values that vary over time.



NOTES: This figure shows the results from estimating Equation (3) for consumer prices when we include category–origin and category–time fixed effects. We include only continuing products in the estimation and weight observations by pre-depreciation expenditure shares. The dots indicate the point estimate and the whiskers are 95% confidence intervals based on standard errors that are clustered at the category–origin level.

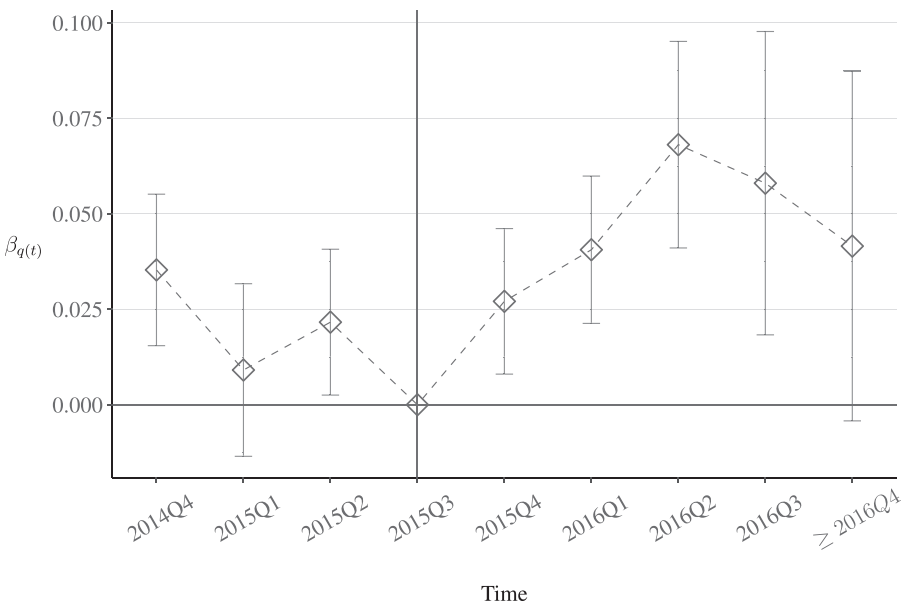
FIGURE 5

DIFFERENCE-IN-DIFFERENCE: CONSUMER PRICES

before the depreciation and increase in the relative price of foreign varieties after the depreciation is also consistent with the appreciation–depreciation cycle of the KZT relative to the RUB (see Figure 2b). Second, we focus on nondurable items that are frequently restocked. Therefore, it is unlikely that the appreciation of the KZT before the depreciation affected the post-depreciation adjustment.

We find that the increase in relative consumer prices of foreign varieties relative to local varieties was small. In the baseline specification, the change in relative consumer prices peaks at a mere 3% three quarters after the depreciation and is not statistically significant at the 95% level from that point onward. Figure A.11 and Table A.17 show that this result is robust across different specifications. First, the results are robust to replacing the category–origin fixed effects with more detailed product variety fixed effects. In this specification, relative foreign prices increase by around 3% throughout the first four quarters after the depreciation and reach their peak at around 5% in the quarters thereafter. Second, the baseline specification treats the price evolution of foreign varieties relative to local alternatives similarly across the two store locations. This is problematic if there is heterogeneity in the assortment of local and foreign varieties across locations or if these locations experience different concurrent unobserved shocks. To this end, we interact the category–origin and category–time fixed effects with store fixed effects. In this case, we find that relative foreign prices increase by around 5.5% shortly after the shock and remain 4.25% higher after four quarters. Finally, relative foreign prices increase to around 6% after four quarters when we account for variety–store and category–time–store fixed effects simultaneously. However, the relative price increase remains small in magnitude compared to the size of the depreciation.

Marginal costs Next, we unpack the relative price adjustment by separately looking at how relative marginal costs and retail markups changed. Figure 6 shows that the relative marginal cost of foreign varieties increases by 3% on impact, reaches its peak in the second quarter at



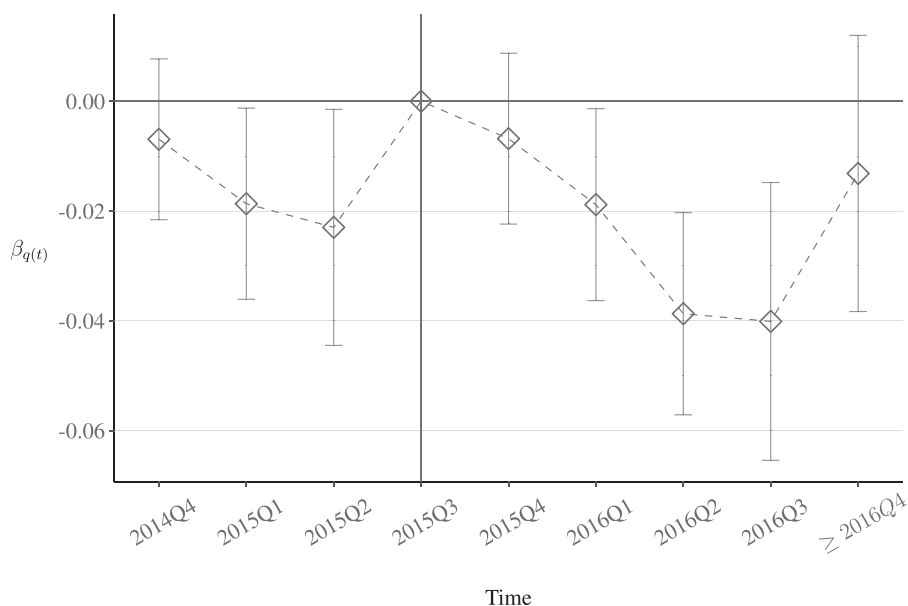
NOTES: This figure shows the results from estimating Equation (3) when we include category–origin and category–time fixed effects. We include only continuing products in the estimation and weight observations by pre-depreciation expenditure shares. The dots indicate the point estimate and the whiskers are 95% confidence intervals based on standard errors that are clustered at the category–origin level.

FIGURE 6
DIFFERENCE-IN-DIFFERENCE: COSTS

about 6.8%, and settles at around 5%. Figure A.12 and Table A.18 show that these results are also robust when we include more flexible fixed effects. Like before, the estimated effects are slightly larger, between 7% and 9% after four quarters, when we control for persistent differences in costs across foreign and local varieties (i.e., variety fixed effects), for store-specific effects (i.e., category–origin–store and category–time–store fixed effects) or for both variety- and store-level effects at the same time.

Several mechanisms can potentially explain this modest relative cost increase. First, in response to the depreciation, exporters could have lowered markups on foreign varieties (Berman et al., 2012; Fitzgerald and Haller, 2014) and manufacturers of local varieties might have increased markups on local varieties (Amiti et al., 2019). Second, sharply rising production costs for local alternatives due to rising nominal wages (see Figure A.5) and more expensive imported intermediate inputs in production (Amiti et al., 2014) are other possible explanations. Third, as explained in Section 2, our cost measure includes the wholesale price paid and distribution costs. If the distribution costs were very large and paid in terms of domestic labor, they would naturally dampen changes in relative costs. Unfortunately, our data do not report the wholesale price and the distribution costs separately. A final reason could be measurement error in the determination of local versus foreign varieties. There are two reasons why measurement error might not be a big concern. On the one hand, the classification works very well for a category in which varieties are differentiated by origin. On the other hand, if measurement error was important, it would imply that our results are a lower bound on the relative adjustment in final consumer prices, costs, and retail markups. Hence, our results would qualitatively not be overturned.

Retail markups Figure 7 illustrates that retail markups fell for foreign varieties relative to local ones and thus counteracted the relative cost increase experienced by foreign varieties. In our baseline specification, we find that four quarters after the depreciation markups on



NOTES: This figure shows the results from estimating Equation (3) for markups when we include category–origin and category–time fixed effects. We include only continuing products in the estimation and weight observations by pre-depreciation expenditure shares. The median pre-depreciation retail markup was 1.13. The dots indicate the point estimate and the whiskers are 95% confidence intervals based on standard errors that are clustered at the category–origin level.

FIGURE 7

DIFFERENCE-IN-DIFFERENCE: MARKUPS

foreign varieties fell by around 4% relative to local varieties. While relative markups are insignificantly different from zero on impact, they gradually drop and reach their trough after four quarters. Retail markups on foreign and local varieties are not different statistically anymore four quarters after the depreciation, which explains the jump in relative prices over this same period. Figure A.13 and Table A.19 show that these results are both qualitatively and quantitatively very robust to including alternative and more detailed sets of fixed effects that account for persistent differences across varieties and locations and for location-specific time effects within product categories.

To understand the decrease in retail markups on foreign varieties relative to local varieties, we distinguish between three possible mechanisms. First, in response to competitive pressure from local stores with fewer foreign products, the retailer might have reduced markups on foreign varieties. However, using more aggregated data from AC Nielsen,²⁶ we show in Table A.10 that prices of food and nonalcoholic beverages did not change differently in small and large stores after the depreciation.²⁷ In addition, Figure A.3 plots the expenditures shares across different store types and illustrates that these shares remained very stable after the depreciation. Both elements point to a relatively stable retail market, making it unlikely that this is the main force acting on relative markups.

Second, the reduction in retail markups for foreign varieties could alternatively be rationalized in demand systems that provide a link between the elasticity of demand and real

²⁶ The data from AC Nielsen cover a much broader sample of stores that sell food and nonalcoholic beverages but provide information on only the 40 best-selling products in each category.

²⁷ Columns (2) and (3) of Table A.10 show the results from a difference-in-difference estimation where we compare the price evolution in large stores relative to small stores for the same category and the same products, respectively. These regressions yield a precisely estimated zero effect.

income.²⁸ However, if a decrease in real income was responsible for the reduction in retail markups on foreign varieties, we should see the same for retail markups on local varieties. In contrast, the data show that, on average, retail markups on local varieties slightly increased.

We argue that the fall in retail markups on foreign varieties relative to local varieties is consistent with optimal pricing by a multiproduct retailer that faces oligopolistic competition and heterogeneous consumers, such as in a mixed-CES demand.²⁹ First, the assumption of oligopolistic competition is necessary because the combination of CES demand with perfect or monopolistic competition leads to constant markups. Second, heterogeneity in consumer preferences is necessary as well. This is because in a CES-demand system without consumer heterogeneity substitution between products only depends on parameters and the market shares of those respective products. Importantly, when products are sold by the same firm, the relevant market share is the firm-level market share (Hottman et al., 2016). Hence, in this case, relative markups on products sold by the same firm are unaffected by changes in relative costs. In contrast, in a CES-demand system with consumer heterogeneity, that is, consumer-specific taste or elasticities of substitution, substitution between products is governed by a weighted average of consumer-specific elasticities of substitution with weights that depend on consumer-specific consumption levels. Consequently, when the composition of consumer-specific consumption levels changes in response to changes in relative costs, variety-level elasticities of demand and, thus, equilibrium markups change accordingly.

Crucially, our data also point toward important heterogeneity in consumer preferences across rich and poor consumers. First, Subsection 3.2 shows that rich consumers spent significantly more on foreign varieties even though they face the same consumer prices, which is evidence of a higher taste for foreign varieties. Second, in Section 5, we show that rich consumers have significantly lower elasticities of substitution compared to poor consumers. This heterogeneity in price sensitivity is a second source of consumer heterogeneity that supports an interpretation of the data through the lens of a mixed-CES preference system.

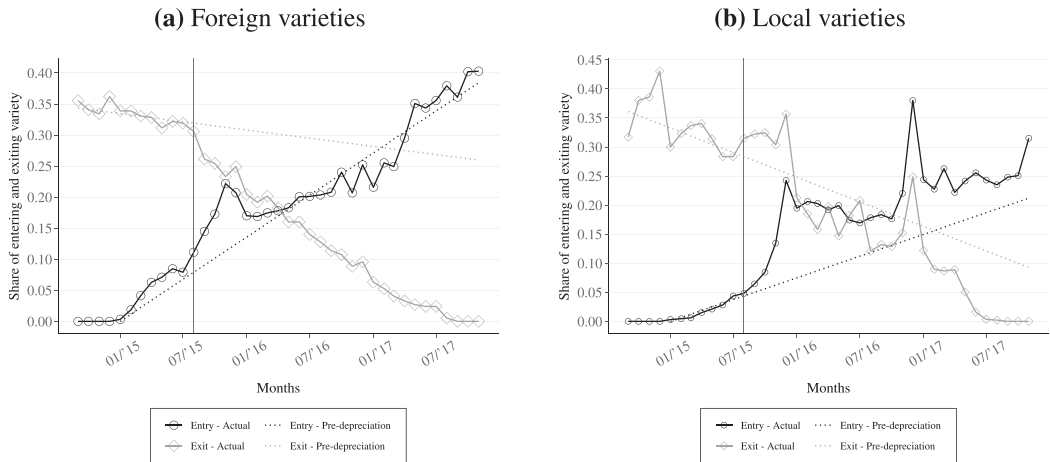
3.4. Changes in Product Availability. Besides changes in final consumer prices, marginal costs, and retail markups, the depreciation also induced substantial adjustments in the set of varieties that were available to consumers. To see this, Table A.20 shows for different subcategories the share of entering, exiting, and continuing varieties in terms of the number of varieties and the expenditure share allocated to them. For now, we define continuing varieties as varieties that were present before the depreciation and were still present in the sample one year after the depreciation. Exiting products are products that were present before the depreciation, but were not present anymore after one year after the depreciation. Entering products were not present before the depreciation, but entered within one year after the depreciation. Across virtually all product categories, the share of continuing varieties in terms of number of varieties is below 50% and is around 70% in terms of expenditure.

However, nontrivial product churning, that is, simultaneous entry and exit of varieties, happens even in more tranquil environments (Bernard et al., 2010; Broda and Weinstein, 2010; Argente et al., 2024).³⁰ This begs the question of whether such changes arose because there was a structural break around the depreciation or because of product churning that would have happened regardless of the depreciation. To discriminate between both explanations, we follow Kehoe and Ruhl (2013) and plot in Figure 8 the expenditure on entering (exiting) varieties, defined as product varieties not available in the first (last) three months of the sample period, as a share of total expenditure over time. As varieties progressively enter (exit) the

²⁸ This could happen through an increase in search intensity (Stroebe and Vavra, 2019; Sangani, 2023) or through changes in the marginal utility of consumption (Mongey and Waugh, 2024)

²⁹ This way of modeling the retail sector is very standard in the industrial organization literature (Hellerstein, 2008; Crawford and Yurukoglu, 2012; Miller and Weinberg, 2017).

³⁰ Steady-state product churning can be quite substantial. In particular, Argente et al. (2024) show that, on average, 12% of sales is accounted for by products that were introduced in the same year and that sales of existing products decrease by around 30% per year.



NOTES: This figure plots the share spent on entering varieties and exiting varieties separately for foreign and local varieties. The trend lines are computed as a linear extrapolation from the pre-depreciation trend in the shares.

FIGURE 8

EXTENSIVE MARGIN

sample the expenditure share on this group of varieties gradually increases (decreases). Yet, if continuous product churning was behind changes at the extensive margin, the reallocation of expenditure toward/away from these varieties would not have changed markedly around the depreciation and would have followed the pre-depreciation trend lines. However, we find that after the depreciation increasingly more expenditure was reallocated away from foreign varieties that eventually exited. Also, right after the depreciation, increasingly more expenditure was allocated to entering local varieties. These patterns align well with workhorse models such as Melitz (2003) and Bernard et al. (2011) in which foreign firms or varieties exit and domestic firms or varieties enter following an increase in trade barriers. As the depreciation permanently realigned the nominal exchange rate closer to the underlying economic fundamentals of the Kazakh economy, a de facto increase in trade barriers occurred, inducing increased entry of local varieties and exit of foreign varieties.

4. FRAMEWORK

4.1. Conceptual Approach. To estimate the impact of the depreciation on consumers' welfare, we compute the compensating variation which is the hypothetical income required to keep a consumer's utility unchanged after being subjected to the depreciation:

$$CV^h = \underbrace{e(\mathbf{P}^1, u_0^h) - e(\mathbf{P}^0, u_0^h)}_{\text{Cost of living effect (CLE)}} + \underbrace{y_1^h - y_0^h}_{\text{Nominal income effect (NI)}},$$

where $\mathbf{P}^t$ is the price vector, u^h is the utility of consumer h , and $e(\cdot)$ is the unit expenditure function. The compensating variation of consumer h depends on how her nominal income, y^h , changed and how her cost of living, $e(\mathbf{P}, u^h)$ changed. As we do not observe the nominal income of consumers, we abstract from changes in the nominal income of households and focus on how the cost of living changed.

To obtain a closed-form solution for $e(\mathbf{P}, u^h)$, we model consumer preferences as a nested mixed-CES demand system for two reasons. First, as explained in the previous section, by allowing for heterogeneity in consumer preferences, the mixed-CES demand system features variable elasticities of demand and generates the empirical facts documented in Section 3.

Second, the CES framework is the workhorse framework to quantify the welfare effects of changes in the set of available varieties. Doing so, we follow Atkin et al. (2018), Jaravel (2019), and Argente and Lee (2021) and capture heterogeneity in consumer preferences by allowing budget shares and elasticities of substitution to vary nonparametrically between income groups whereas households within an income group share the same preferences.

4.2. Preferences. There are four levels of aggregation in the data: (i) the category level (e.g., Food), (ii) the subcategory level (e.g., Fruit), (iii) the product group level (e.g., Stone-fruit), and (iv) the product level (e.g., Peaches).³¹ Each product comes in different varieties, which can be local or foreign. We assume that substitution occurs along the following two nests:³² The upper nest captures substitution across products (e.g., rice vs. bread) and the lower nest substitution across varieties within the same category (e.g., Basmati rice vs. Jasmin rice). We choose to define the upper nest at the product level for two reasons. First, we want to allow for substitution between products after the depreciation. For instance, if the depreciation causes bread prices to increase relative to rice, consumers might choose to substitute bread for rice. The degree of substitution is governed by the elasticity of substitution σ_c^h .³³ Second, the product level is the finest level of aggregation without meaningful product entry or exit after the depreciation. There are products that either enter or exit after the depreciation, but their sales share in total category spending is around 0.1%. The lowest nest is defined at the variety level where the elasticities, η_p^h , govern how consumers substitute between varieties of the same product. More formally, the aggregator in the upper nest is given by³⁴:

$$U_t^h = \left[\sum_{p \in \Omega} \xi_p^h Q_{p,t}^h \frac{\sigma^h}{\sigma^h - 1} \right]^{\frac{\sigma^h - 1}{\sigma^h}},$$

where $Q_{p,t}^h$ is the aggregate consumption of product p by households in income group h at time t , ξ_p^h is a product level demand shifter, which may differ across income groups, Ω is the set of available products and σ^h is the elasticity of substitution. The product-specific quantity, $Q_{p,t}^h$ is a CES aggregator over individual varieties:

$$Q_{p,t}^h = \left[\sum_{i \in \Omega_{p,t}} \xi_{pi}^h Q_{pi,t}^h \frac{\eta_p^h}{\eta_p^h - 1} \right]^{\frac{\eta_p^h - 1}{\eta_p^h}},$$

where $Q_{pi,t}^h$ is the consumption of a variety i by households in income group h at time t , ξ_{pi}^h is a variety-level demand shifter, $\Omega_{p,t}$ is the set of varieties available at time t within product p and η_p^h is the elasticity of substitution between varieties. In this way, consumers of the same income group have the same elasticity of substitution across foreign and local varieties. Local and foreign varieties directly compete within highly detailed product categories and face the same elasticity of substitution within income groups.³⁵ At the same time, consumer taste

³¹ To compare the level of aggregation with the widely used Nielsen HomeScan database (Hottman et al., 2016; Jaravel, 2019; Argente and Lee, 2021), this data set contains 184 different product groups and 900 different products which are comparable in their level of aggregation to the “Product Groups” and “Product Modules” in the Nielsen HomeScan database.

³² We execute the analysis for one category, that is, food and nonalcoholic beverages. We are silent on the way we aggregate across categories. For instance, a Cobb–Douglas aggregator would be suitable.

³³ The model captures this type of substitution through the product level Sato–Vartia weight which is constructed from the pre-and post-depreciation expenditure shares.

³⁴ For notational simplicity, we drop the category index c .

³⁵ This is in contrast to international trade applications where interactions between foreign and local goods happen at a higher level of aggregation (see, for instance, Schmitt-Grohé and Uribe, 2018; Fajgelbaum et al., 2020).

is allowed to differ between foreign and local varieties and between consumers of different income groups. In this way, the preference structure rationalizes differences in expenditure shares between foreign and local varieties and rich and poor consumers. Given this structure, the category-level and product-level unit expenditure functions are given by:

$$P_t^h = \left[\sum_{p \in \Omega} \xi_p^h P_{p,t}^h \right]^{\frac{1}{1-\sigma^h}}, \quad P_{p,t}^h = \left[\sum_{i \in \Omega_{p,t}} \xi_{pi,t}^h P_{pi,t}^h \right]^{\frac{1}{1-\eta_p^h}},$$

where $P_{pi,t}^h$ is the price of variety i at time t .

4.3. Price Index Decomposition. Because the income-group-specific utility functions are homothetic, the change in the cost of living coincides with the change in the unit expenditure function:

$$\frac{CLE_t^h}{e(P^0, u_0^h)} = \frac{e(P^t, u_0^h)}{e(P^{t-1}, u_0^h)} - 1 = \prod_{p \in \Omega} \left[\frac{P_{p,t}^h}{P_{p,t-1}^h} \right]^{\omega_{p,t}^h} - 1,$$

where $\omega_{p,t}^h$ are the Sato–Vartia weights which are given by:

$$\omega_{p,t}^h \equiv \frac{\frac{\phi_{p,t}^h - \phi_{p,t-1}^h}{\ln \phi_{p,t}^h - \ln \phi_{p,t-1}^h}}{\sum_{p \in \Omega} \frac{\phi_{p,t}^h - \phi_{p,t-1}^h}{\ln \phi_{p,t}^h - \ln \phi_{p,t-1}^h}}, \quad \phi_{p,t}^h \equiv \frac{\sum_{i \in \Omega_{p,t}} P_{pi,t} \cdot Q_{pi,t}^h}{\sum_{p \in \Omega} \sum_{i \in \Omega_{p,t}} P_{pi,t} \cdot Q_{pi,t}^h}.$$

The change in the product-level unit expenditure function can be further decomposed into a term that depends on changes in variety-level prices for continuing products and a term that captures changes in product availability at the variety level (Feenstra, 1994):

$$\frac{CLE_t^h}{e(P^0, u_0^h)} = \prod_{p \in \Omega} \left[\prod_{i \in \Omega_p^{t-1 \cap t}} \left(\frac{P_{pi,t}}{P_{pi,t-1}} \right)^{\omega_{pi,t}^h} \cdot \left(\frac{\lambda_{p,t}^h}{\lambda_{p,t-1}^h} \right)^{\frac{1}{\eta_p^h - 1}} \right]^{\omega_{p,t}^h} - 1,$$

where $\omega_{pi,t}^h$ are the variety-level Sato–Vartia weights given by:

$$\omega_{pi,t}^h \equiv \frac{\frac{\phi_{pi,t}^h - \phi_{pi,t-1}^h}{\ln \phi_{pi,t}^h - \ln \phi_{pi,t-1}^h}}{\sum_{i \in \Omega_p^{t-1 \cap t}} \frac{\phi_{pi,t}^h - \phi_{pi,t-1}^h}{\ln \phi_{pi,t}^h - \ln \phi_{pi,t-1}^h}}, \quad \phi_{pi,t}^h \equiv \frac{P_{pi,t} \cdot Q_{pi,t}^h}{\sum_{i \in \Omega_p^{t-1 \cap t}} P_{pi,t} \cdot Q_{pi,t}^h}$$

and $\phi_{pi,t}^h$ is the expenditure share of variety i at time t and $\Omega_p^{t-1 \cap t} \equiv \Omega_{p,t-1} \cap \Omega_{p,t}$. The variety effect is defined as the ratio of the expenditure share of continuing products relative to all available varieties at time t and the expenditure share of continuing products relative to all available varieties at time $t - 1$:

$$\frac{\lambda_{p,t}^h}{\lambda_{p,t-1}^h} \equiv \frac{\frac{\sum_{i \in \Omega_p^{t-1 \cap t}} P_{pi,t} \cdot Q_{pi,t}^h}{\sum_{i \in \Omega_{p,t}} P_{pi,t} \cdot Q_{pi,t}^h}}{\frac{\sum_{i \in \Omega_p^{t-1 \cap t}} P_{pi,t-1} \cdot Q_{pi,t-1}^h}{\sum_{i \in \Omega_{p,t-1}} P_{pi,t-1} \cdot Q_{pi,t-1}^h}}$$

weighted by the elasticity of substitution. Finally, using the definition of retail markups as the ratio of the final consumer price and the marginal cost, $P_{pi,t} \equiv M_{pi,t} \cdot C_{pi,t}$, we arrive at the final decomposition:

$$(4) \quad \frac{CLE_t^h}{e(\mathbf{P}^0, u_0^h)} = \prod_{p \in \Omega_p} \left[\underbrace{\prod_{i \in \Omega_p^{t-1 \cap \tau}} \left(\frac{C_{pi,t}}{C_{pi,t-1}} \right)^{w_{pi,t-1}^h} \cdot \prod_{i \in \Omega_p^{t-1 \cap \tau}} \left(\frac{M_{pi,t}}{M_{pi,t-1}} \right)^{w_{pi,t-1}^h}}_{\text{Price channel}} \cdot \underbrace{\prod_{i \in \Omega_p^{t-1 \cap \tau}} \left(\frac{P_{pi,t}}{P_{pi,t-1}} \right)^{\omega_{pi,t}^h - w_{pi,t-1}^h} \cdot \left(\frac{\lambda_{p,t}^h}{\lambda_{p,t-1}^h} \right)^{\frac{1}{\eta_p^h - 1}}}_{\text{Substitution channel} \cdot \text{Variety channel}} \right]^{\omega_{p,t}^h} - 1.$$

This shows that we can decompose the cost of living of consumer h into three different channels: (i) the price channel, which further decomposes into a cost and markup effect, (ii) the substitution channel, and finally (iii) the variety channel.

The price channel is defined as the covariance between changes in consumer prices and pre-depreciation expenditure shares of continuing varieties. It gives rise to distributional effects if different income groups have different expenditure shares on different varieties. For example, if rich consumers spend relatively more on foreign varieties and if foreign varieties experience a greater price increase, then the price channel will increase more for rich consumers. By further decomposing the price channel into a cost and markup effect, we provide more insight into the transmission of the depreciation into final consumer prices and move beyond previous work that mostly focuses on models with constant markups (Fajgelbaum and Khandelwal, 2016; Borusyak and Jaravel, 2021). Since rich consumers had larger expenditure shares on foreign varieties before the depreciation, we expect the cost channel to increase the cost of living relatively more for rich consumers and the markup channel to offset this increase as markups of foreign varieties fell relative to local varieties.

The substitution channel is defined as the covariance between changes in final consumer prices and the difference in the variety-level Sato–Vartia weights and variety-level pre-depreciation expenditure shares. Intuitively, if consumers reallocate expenditure away from varieties with higher price increases, their ex post cost of living will be lower. If low-income consumers have higher elasticities of substitution, we expect stronger substitution away from varieties with higher price increases for poor consumers, attenuating their increase in the cost of living relative to rich consumers.³⁶

The variety channel quantifies how changes at the extensive margin translate into changes in the cost of living. It is composed of the ratio of the expenditure share on continuing varieties after and before the depreciation and the variety-level elasticity of substitution. If the ratio of expenditure shares is below one, the appeal of the varieties that entered must have been greater compared to the appeal of varieties that exited. The extent to which this reallocation of expenditure translates into changes in the cost of living depends on the elasticity of substitution. When the elasticity is high, products are perceived as good substitutes, and adding new varieties increases utility only marginally. Whether the variety effect will have distributional cost-of-living effects depends on the difference in the elasticity of substitution and

³⁶ As the substitution channel captures differences in the way rich and poor consumers substitute in response to the depreciation, it corresponds to the unequal expenditure switching channel emphasized in Auer et al. (2023).

the observed reallocation of expenditure from exiting to entering varieties between rich and poor consumers.

5. QUANTIFYING CHANGES IN THE COST OF LIVING

To estimate how the cost of living changed, we first need to estimate the variety-level elasticities of substitution. We start by discussing our empirical strategy. Hereafter, we present both average and income-group-specific elasticities of substitution. Finally, we document how the cost of living changed following the depreciation and how rich and poor consumers were differently affected.

5.1. Estimating the Elasticities of Substitution. Applying Shephard's lemma to the product-level unit expenditure function, we obtain the residual demand for variety i :

$$(5) \quad Q_{pi,t}^h = \left(\xi_{pi,t}^h \right)^{\eta_p^h - 1} \left(\frac{P_{pi,t}^h}{P_{p,t}^h} \right)^{-\eta_p^h} Q_{p,t}^h.$$

After taking logs, Equation (5) becomes:

$$q_{pi,t}^h = -\eta_p^h p_{pi,t}^h + \eta_p^h p_{p,t}^h + q_{p,t}^h + \left(\eta_p^h - 1 \right) \ln(\xi_{pi,t}^h).$$

where lowercase letters indicate logarithmic transformations. The parameters of interest are the elasticities of substitution η_p^h . To consistently estimate the elasticity, we need to overcome a set of econometric challenges. First, the product-level price index $P_{p,t}^h$ and quantity index $Q_{p,t}^h$ depend on the demand shifters and are unobserved. To deal with this simultaneity issue, we flexibly account for them using product–income–time fixed effects.³⁷ Second, the demand shifters, $\xi_{pi,t}^h$, are also unobserved. If firms have prior knowledge of $\xi_{pi,t}^h$, prices are likely set with $\xi_{pi,t}^h$ in mind. To overcome this issue, we decompose the demand shifters into a variety–income group fixed effect and a residual term as follows: $\ln(\xi_{pi,t}^h) = \theta_{pi}^h + \varepsilon_{pi,t}^h$. In this way, our estimating equation becomes:

$$(6) \quad q_{pi,st}^h = \beta_p^h p_{pi,st}^h + \theta_{pi}^h + \lambda_{p,t}^h + \varepsilon_{pi,st}^h,$$

where we have added the subscript s to indicate that we observe prices and quantities at the variety–store–income group–time level. While the category–income–time fixed effects, $\lambda_{p,t}^h$, take care of the simultaneity bias, there may still be unobserved time-varying demand shifters that are correlated with prices. To address this concern, we exploit the spatial dimension of our data. We use the price of variety i in location s' as an instrument for the price of the same variety in location s . We believe this is an appropriate instrument as Metro is using near uniform pricing across the two stores in Almaty and Astana.³⁸ More specifically, Figures A.14 and A.15 replicate some of the evidence presented in Dellavigna and Gentzkow (2019) and indicate that across the two stores prices of the same variety tend to be highly correlated in both the cross section and in the time dimension. By employing uniform pricing across locations, it is much more likely that retailers adjust prices in response to common shocks, for example, changes in costs, than in response to idiosyncratic local demand shocks. Because we study a period in which the depreciation of the Tenge induced large changes in the costs of both local and foreign varieties, price variation in our sample likely stems from such a common cost shock.

³⁷ See Atkin et al. (2018), Arkolakis et al. (2019), and Faber and Fally (2022) for a similar strategy.

³⁸ This identification strategy was first introduced in Hausman (1996) and was subsequently used by Nevo (2001), Dellavigna and Gentzkow (2019), and Faber and Fally (2022).

TABLE 2
ELASTICITY OF SUBSTITUTION: AGGREGATE

	OLS				IV			
$q_{i,p,t}$	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$P_{pi,st}$	-2.24*** (0.133)	-2.15*** (0.133)	-1.43*** (0.068)	-1.32*** (0.071)	-3.17*** (0.214)	-3.08*** (0.224)	-1.55*** (0.077)	-1.41*** (0.085)
Product × Quarter FE	✓				✓			
Product × Month FE		✓				✓		
Product × Quarter × Store FE			✓				✓	
Product × Month × Store FE				✓				✓
Variety FE	✓	✓			✓	✓		
Variety × Store FE			✓	✓			✓	✓
First-stage F -stat	—	—	—	—	516.8	348.0	6778.3	5823.4
R^2	0.056	0.070	0.055	0.071	—	—	—	—
Nr. obs	769,717	769,717	769,717	769,717	620,806	620,806	620,806	620,806

NOTE: This table shows the estimates of the elasticities of substitution pooled across product categories and pooled across consumers. Columns (1)–(4) are OLS estimates and columns (5)–(8) are estimated using the Hausman instrument as the instrument. The first-stage F -statistic refers to the effective first-stage F -statistic developed by Montiel-Olea and Pflueger (2013) which is valid under non-i.i.d. distributed errors. Standard errors are reported below the coefficient in brackets and are clustered at the store-product level. Significance is at the *10%, **5%, and ***1% level.

Nevertheless, the crucial assumption underpinning the consistent estimation of the elasticities is that the Hausman instrument is not contaminated by common demand shocks. Two possible common cost shocks come to mind. First, if the devaluation also induced changes in real income and shifted relative demand across varieties, for example, by affecting the demand for foreign varieties relatively more, the elasticities might be inconsistently estimated. Whereas we cannot account for variety-specific shifts over time, we consider specifications that control for shifts between local and foreign varieties within product categories below. Second, the estimation strategy also rules out demand shocks driven by national advertising. To partially address this concern, we follow Dellavigna and Gentzkow (2019) and show that the results are robust to controlling for seasonal variety-specific demand shocks.

5.2. Estimates of the Elasticity of Substitution. To estimate the elasticities of substitution, we regress monthly purchased quantities on consumer prices (inclusive of sales and coupons) and we include all periods. Also, we include all varieties in the estimation, so we do not distinguish between continuing, entering, and exiting varieties here. We report unweighted regressions and cluster standard errors at the category–store level. We first provide estimates that average across income groups and then estimate elasticities of substitution for each income group separately.

Average elasticity of substitution Columns (1)–(4) of Table 2 present the ordinary least square (OLS) estimates of the average elasticity of substitution and columns (5)–(8) present the instrumental variable (IV) estimates. Column (1) shows the result when we include product–quarter fixed effects, which filter out the price and quantity indices, and variety fixed effects to account for the demand shifters. In this setup, we recover a negative and statistically significant estimate of -2.24 . When we replace the product–quarter fixed effects with more detailed product–month fixed effects in column (2), we find that the elasticity is almost the same and is estimated at -2.15 . When we allow the demand shifters and price indices to also differ across locations in columns (3) and (4), the estimated elasticities decrease to -1.43 and -1.31 , respectively, but remain below the theoretical constraint of -1 .

Looking at columns (5)–(8), we first note that the Hausman-type instrument is strong as the first-stage F -statistics are always substantially above the conventional critical values, which is consistent with the presence of near uniform pricing. Second, the IV estimates are statistically

TABLE 3
ELASTICITY OF SUBSTITUTION: PER INCOME GROUP (20–80% SPLIT)

$q_{i,p,t}$	OLS				IV			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$p_{i,p,t} \cdot 1$ (Poor)	−3.65*** (0.243)	−3.51*** (0.252)	−2.35*** (0.146)	−2.27*** (0.160)	−5.2*** (0.467)	−5.18*** (0.536)	−2.51*** (0.184)	−2.53*** (0.215)
First-stage F -stat	—	—	—	—	292.6	170.8	2,920.1	2,213.8
R^2	0.034	0.054	0.018	0.043	—	—	—	—
Nr. obs	190,063	190,063	190,063	190,063	151,759	151,759	151,759	151,759
$p_{i,p,t} \cdot 1$ (Middle)	−2.34*** (0.139)	−2.29*** (0.145)	−1.57*** (0.077)	−1.43*** (0.083)	−3.17*** (0.244)	−3.15*** (0.274)	−1.7*** (0.100)	−1.52*** (0.113)
First-stage F -stat	—	—	—	—	619.2	415.3	7,571.5	6,232.5
R^2	0.104	0.120	0.100	0.120	—	—	—	—
Nr. obs	329,014	329,014	329,014	329,014	264,839	264,839	264,839	264,839
$p_{i,p,t} \cdot 1$ (Rich)	−1.24*** (0.155)	−1.14*** (0.158)	−0.964*** (0.089)	−0.825*** (0.095)	−2.22*** (0.263)	−2.17*** (0.288)	−1.22*** (0.106)	−1.06*** (0.115)
First-stage F -stat	—	—	—	—	443.7	295.3	5413.4	4578.2
R^2	0.052	0.069	0.045	0.066	—	—	—	—
Nr. obs	250,640	250,640	250,640	250,640	204,208	204,208	204,208	204,208
Product $\times$ Quarter FE	✓				✓			
Product $\times$ Month FE		✓				✓		
Product $\times$ Quarter $\times$ Store FE			✓				✓	
Product $\times$ Month $\times$ Store FE				✓				✓
Variety FE	✓	✓			✓	✓		
Variety $\times$ Store FE			✓	✓			✓	✓

NOTE: This table shows the estimates of the elasticities of substitution for each income group separately, but pooled across product categories. The results per income group are obtained by estimating Equation (6) separately for each income group. Panel (a) shows the results for the **relatively low-income group, panel (c) for the relatively high-income group, and panel (b) for consumers classified in the middle-income group. Columns (1)–(4) are OLS estimates and columns (5)–(8) are IV estimates using the Hausman instrument. The first-stage F -statistic refers to the effective first-stage F -statistic developed by Montiel-Olea and Pflueger (2013) which is valid under non-i.i.d. distributed errors. Standard errors are reported below the coefficient in brackets and are clustered at the store-product level. Significance is at the *10%, **5%, and ***1% level.

significant and deliver more elastic demand curves relative to the OLS counterparts. For example, in the most basic setup, the IV estimate is -3.17 whereas the corresponding OLS estimate is -2.24 . Third, the estimates reported in columns (5)–(8) are well in the range of previous estimates in the literature. Using a similar empirical strategy, Dellavigna and Gentzkow (2019) estimate elasticities of demand for similar product categories sold by U.S. retailers and report an average elasticity of substitution for food products of around -2.8 . These estimates are also close to own-price elasticities for food products that have been reported in the discrete-choice literature: Nevo (2001) estimate own-price elasticities of demand for breakfast cereal that range between -2.34 and -4.25 and Hendel and Nevo (2013) find own-price elasticities of demand between -2.46 and -2.94 for soft drinks.

Consumer heterogeneity To understand whether consumers of different income groups might have different elasticities of substitution, we reestimate Equation (6) separately for each income group. We estimate one regression for each income group to ensure that the fixed effects vary at the income group level and that they filter out income group-specific price indices and variety-level demand shifters. Table 3 reports the results for the same fixed effect specifications as Table 2 and shows the results for the relatively low-, middle-, and high-income groups, respectively.

Table 3 shows that the instruments remain strong and that the IV estimates are statistically significant and negatively estimated. Also, the IV estimates again deliver more elastic demand curves relative to the corresponding OLS estimates. More importantly, regardless of the specification, we find that rich consumers have lower elasticities of substitution compared to poor

consumers. This indicates that rich consumers consider the alternatives in their choice set as less substitutable and will value changes in product variety more relative to poor consumers. The finding that high-income consumers have lower elasticities of substitutions is in line with a substantial IO literature (e.g., Berry et al., 1995 for cars; Nevo, 2001 for breakfast cereal; and Dellavigna and Gentzkow, 2019 groceries). Also, they are quantitatively in line with the differences in elasticities of substitution across income groups reported in Handbury (2021) and Auer et al. (2023). For instance, Auer et al. (2023) find that a tripling of household income leads to a fall in the elasticity of substitution between 2.12 and 2.42.

Robustness We consider four robustness checks. First, we consider the possibility that the depreciation not only acted as a cost shock but also changed relative demand across foreign and local varieties through its effect on real income. To investigate this, we follow Bems and Giovanni (2016) and consider a specification in which we interact the category–time fixed effects with origin fixed effects. In this way, we account for a common change in the demand for foreign varieties relative to local varieties. We find that the elasticities are still precisely estimated and are slightly more inelastic compared to the baseline estimates (see Table A.22). Crucially, the difference between the elasticities for rich and poor consumers remains quantitatively important.

Second, the baseline estimates do not control for changes in variety-level demand over time that are common between the two store locations. To assess the robustness of the results in controlling for seasonal demand shocks, we follow Dellavigna and Gentzkow (2019) and consider a specification in which we replace the variety–store fixed effects with variety–store–year and variety–store–month-of-the-year fixed effects. In this way, we control for seasonal variety-specific demand shocks that are common across stores and are allowed to differ every year. Tables A.23 and A.29 show that the estimates we recover are very close to the ones we recover with only variety–store fixed effects, both on average and between groups of consumers.

Third, we also consider clustering the standard errors at the monthly level to account for systematic cross-sectional correlation across varieties induced by the depreciation. Tables A.21 and A.27 show that the elasticities remain precisely estimated.

Finally, when we classify consumers according to their total expenditure, we find that the difference between the elasticities of substitution of rich and poor consumers is both qualitatively and quantitatively preserved (see Table A.30).

5.3. Cost-of-Living Effects. This section presents how the cost of living changed in the four quarters following the depreciation. To have a sense of the average change in the cost of living, we start with the aggregate results. Hereafter, we show that the change in the cost of living differed between rich and poor consumers.

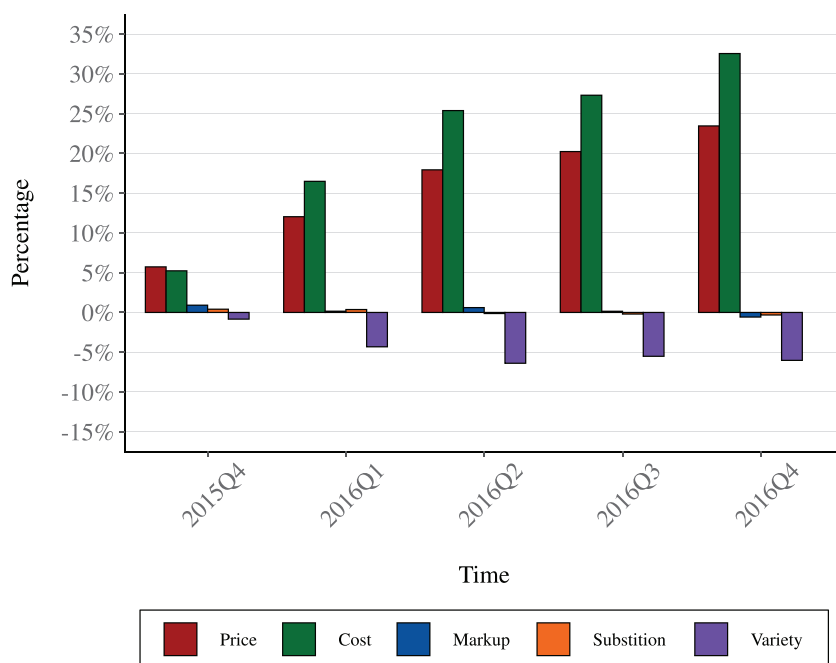
Aggregate effects To obtain the aggregate decomposition results, we first compute each of the four components: (i) cost channel, (ii) markup channel, (iii) substitution channel, and (iv) product variety channel separately and combine them to obtain the overall cost-of-living effect.³⁹ To calculate the components, we fix period $t - 1$ to be equal to the pre-depreciation quarter and compute for each of the ensuing quarters the cumulative difference relative to the pre-depreciation.⁴⁰

Figure 9 shows the average change in the cost of living and the decomposition into the different channels.⁴¹ Figure 9 clearly shows that the cost of living went up considerably after the depreciation. After one year, the cost of living increased by a little under 25%. The transmission of the exchange rate shock into prices was gradual as the cost of living increased by 5%

³⁹ To compute the variety effect, we use the estimate in column (5) of Table 2. We use this estimate as it yields the most conservative results for the variety effect.

⁴⁰ We define the pre-depreciation quarter as June 2015, July 2015, and August 2015. We proceed in this way because the depreciation was on August 20 and Figure 3 indicates that prices did not respond at all in August 2015.

⁴¹ The same results are also displayed in Table A.24.



NOTES: This figure shows the average change in the cost of living and in each of the channels. The results are obtained after pooling across all income groups and estimating the variety effect when we restrict the elasticity of substitution to be the same across all product categories. Doing so, we use the estimate of column (5) in Table 2. These effects are cumulative effects relative to the quarter before the depreciation, which is defined as June, July, and August 2015. The size of each bar is expressed in percentage differences and is obtained by subtracting 1 and multiplying by 100 each of the numbers in Table A.24.

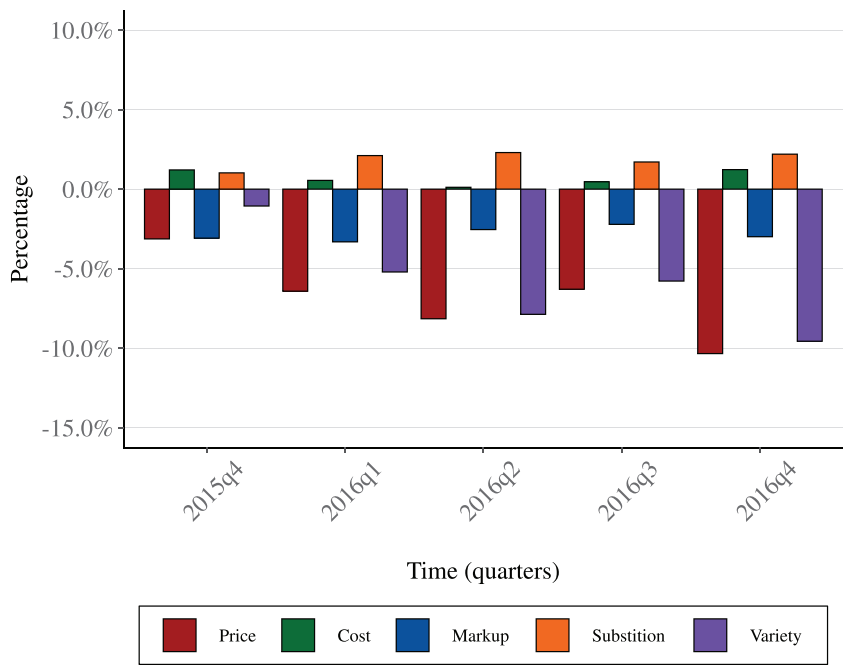
FIGURE 9

COST-OF-LIVING: AGGREGATE EFFECTS

after one quarter and steadily grew to a little under 25% after five quarters. This is qualitatively in line with Figure 3 which also showed that pass-through converged after 12–15 months and is also consistent with other large devaluation episodes as described in Burstein et al. (2005) and Alessandria et al. (2010).

The increase in costs for continuing varieties was the main driver of the aggregate cost-of-living increase. Two quarters after the depreciation, the marginal cost of food and beverages went up by more than 15% and by 28% after four quarters. This corresponds roughly to an aggregate pass-through rate into marginal costs of 40–50% which is quantitatively in the range of estimates provided by Goldberg and Campa (2010) and recent work on dominant currencies in international trade (Gopinath et al., 2020). In the aggregate, the markup and substitution channels did not significantly dampen the increase in the cost of living.

In contrast, four quarters after the depreciation, the variety channel had dampened the increase in the cost of living by about 6%. This implies that the taste-adjusted price of entering varieties was superior to the taste-adjusted price of the varieties that exited. As mentioned before, the fact that the variety effect is quantitatively important is consistent with the closed economy literature on product churning in which relatively less appealing are frequently replaced by relatively more appealing varieties (Bernard et al., 2010; Broda and Weinstein, 2010; Argente et al., 2024). In addition, Argente and Lee (2021) show that the variety channel also had a dampening effect on the cost of living for all U.S. income groups during the Great Recession in the United States by replacing older and less appealing with more appealing varieties. Whereas the effect of changes in product assortment on pass-through into prices has been studied before (Nakamura and Steinsson, 2012; Cavallo et al., 2014; Goetz and



NOTES: This figure shows the distributional cost-of-living results using the quintiles definition of the income distribution. The results are obtained by computing each of the components separately for each income group. The variety effect is computed by allowing the elasticity of substitution to vary across income groups and using the estimates of the elasticity of substitution reported in column (5) of Table 3. These effects are cumulative effects relative to the quarter before the devaluation, which is defined as June, July, and August 2015 and coincide with the ratio column for each channel as displayed in Table A.34. The size of each bar is calculated by subtracting 1 and multiplying by 100 each of the numbers in the corresponding tables.

FIGURE 10
COST-OF-LIVING: DISTRIBUTIONAL EFFECTS—PER INCOME GROUP (20–80% SPLIT)

Rodnyansky, 2023), our results show that accounting for changes in product variety and measuring their welfare impact is important to translate pass-through estimates into welfare terms.

Distributional effects To obtain the distributional effects, we apply the same steps as for the aggregate effects but now with income group-specific expenditure shares and elasticities.⁴² To assess the distributional effects, we compute the ratio of the value for the high-income group (H) relative to the value of the low-income group (L) for the cumulative change in the cost of living and in each of the channels.

Figure 10 presents the distributional effects of the depreciation for the income definition based on quintiles.⁴³ We find that the cost of living went up by less for rich consumers compared to poor consumers as the cost-of-living increase is 5% and up to 10% lower for rich consumers compared to poor consumers two and five quarters after the depreciation, respectively.

Notably, due to offsetting intensive margin channels, the majority of the distributional cost-of-living effect did not arise from changes associated with the set of continuing products. First, the cost and markup channels, which collectively make up the price channel, moved in opposite directions. Depending on the horizon we consider, the baseline results show that the cost

⁴² For the elasticities of substitution, we use the IV estimates displayed in column (5) of Table 3 as they yield conservative results for the variety effect.

⁴³ The same results are also displayed in Table A.34.

channel led to a 0.1–1.2% increase in the cost of living of rich consumers compared to poor consumers. At the same time, the markup channel moved in the opposite direction and attenuated the cost-of-living increase by 2–2.8% for rich consumers. As the cost and markup channels diverged, they made consumer prices go up by less for rich consumers relative to poor consumers. This observation is congruent with the fact that rich consumers have on average a higher expenditure share on foreign varieties whose costs increased more and retail markups declined after the depreciation.

Second, whereas consumer prices on average increased by less for rich consumers, the substitution channel led to a greater cost-of-living increase for rich consumers. Our baseline results point to a relative increase in their cost of living through the substitution channel by 2.1% and 2.3% after two and four quarters, respectively. As the substitution channel measures how intensely consumers substitute away from continuing varieties that become relatively more expensive, this is consistent with the fact that we estimate that poor consumers have higher elasticities of substitution. This is also in line with Bems and Giovanni (2016) and Auer et al. (2023) who find that poor consumers reduced the cost-of-living increase by substituting more toward cheaper continuing products. Taken together, the distributional cost-of-living effect we document did not arise from changes associated with the set of continuing products because the price channel was quantitatively offset by the substitution channel.

Adjustments on the extensive margin, however, were the main driver of the distributional effects of the depreciation on the cost of living. Figure 10 illustrates that rich consumers benefitted more from the changes in variety after the depreciation. After two and four quarters, changes in product variety subdued the cost-of-living increase by 2.5% and 3% for poor consumers and 7.5% and 8.6% for rich consumers. This translates into a cost-of-living inequality of 5.2% after two quarters and 5.7% after four quarters. Despite its quantitative importance in other settings, the extensive margin has not been incorporated before to study the aggregate and distributional welfare consequences of large depreciation. Our results show that accounting for changes in the choice set of consumers is paramount to determine whether relatively rich or relatively poor suffer more from depreciation.

The differences in the variety effect for rich and poor consumers can be driven by two forces: (i) differences in substitution toward new varieties and away from varieties that exited and (ii) differences in the elasticity of substitution. First, to understand differential switching, Figure A.19 shows the distribution of the ratio of the expenditure share on continuing varieties after the depreciation relative to before the depreciation across product categories for rich and poor consumers separately. These ratios measure the extent to which rich and poor consumers substituted toward new varieties and away from varieties that exited. When we test for the equality between these distributions with a parametric and nonparametric test, we are unable to reject the null hypothesis of equal switching. Second, differences in the elasticities of substitution imply that for the same level of switching, the welfare effect will differ. This is because lower elasticities of substitution imply that varieties are perceived to be more differentiated such that the same level of switching implies a larger change in welfare. As we have provided robust evidence that rich consumers have lower elasticities of substitution, heterogeneity in the elasticities of substitution across income groups seems to be the key driver of the distributional effects stemming from the variety effect.

Alternative cutoffs The distributional results that we have presented so far are calculated based on categorizing rich and poor consumers as those consumers whose consumption basket ranks within the top and the bottom quintile in terms of the unit cost. To ensure that our results do not depend on this particular cutoff, we replicate the analysis by considering terciles, quartiles, and deciles as alternative cutoffs. Figures A.17(a)–(d) and Tables A.32–A.35 show that the results are robust across these alternative definitions. Reassuringly, the differences between rich and poor consumers in terms of the cost of living and the individual channels generally grow when we start from the loosest definition, that is, terciles, and move to the strictest definition, that is, deciles.

Heterogeneity across categories Also, the results that we have discussed are based on the assumption that the elasticities of substitution are the same across different product categories. However, if changes in product variety are more concentrated in product categories with higher elasticities of substitution, using homogeneous elasticities could lead to overstating the variety effects. For this reason, we check whether the results are robust to accounting for heterogeneity in the elasticities of substitution.

Starting with the aggregate cost-of-living effects, we reestimate Equation (6) separately for each subcategory and obtain an elasticity of substitution for each of the 14 subcategories.⁴⁴ Table A.25 presents the IV results and shows that the distribution is indeed quite dispersed.⁴⁵ Figure A.16 presents the aggregate cost-of-living effects when we account for heterogeneity in the elasticities of substitution.⁴⁶ Relative to the homogeneous case, the product variety channel strengthens and exerts a greater dampening effect on the aggregate cost-of-living increase. In this setup, changes in the choice set of consumers subdue the rise in the cost of living between 8% and 10% depending on the horizon we look at.

To check the robustness of the distributional effects, we adjust Equation (6) in two ways. First, we not only estimate category-level elasticities of substitution but also interact prices with income group levels. In this way, we estimate 42 different elasticities that vary at the subcategory–income group level. Second, we interact the category–time and variety fixed effects with income group fixed effects to allow for income group-specific price indices, expenditure levels, and demand shifters. We present the estimates in Table A.31. Tables A.36–A.39 and Figures A.18(a)–(d) show the distributional cost-of-living results. First, in line with the aggregate results, accounting for between-category heterogeneity in the elasticities of substitution tends to strengthen the welfare effects of changes in product variety. In particular, for both low-income and high-income consumers, the product variety channel dampened the cost-of-living increase by more. Second, rich consumers still experienced slower growth in the cost of living in the year following the depreciation. In particular, rich consumers experienced a 3%, 5%, and 7% slower increase in their cost of living one, two, and three quarters after the depreciation and this pattern is consistent across the different income group definitions. At the same time, the heterogeneous results are less persistent as the gains from changes in product variety seem to be roughly equal across high-income and low-income consumers after four quarters. Altogether, the heterogeneous results suggest that there was still a substantially lower growth in the cost of living of relatively high-income over the first three quarters after the depreciation but that these growth paths seem to have converged after four quarters.

6. CONCLUSION

This article analyzes the impact of the depreciation of the KZT in August 2015 on consumer prices, costs, and retail markups of local and foreign products and how these adjustments induce distributional cost-of-living effects. To this end, we leverage novel scanner data from a supermarket, Metro, in Kazakhstan at the product and transaction level. The depreciation had a potentially considerable welfare effect as it pushed up the final consumer prices by 25% after one year and induced substantial changes in the set of available products one year after the depreciation.

We document that rich consumers have substantially larger expenditure shares on foreign varieties relative to poor consumers within highly detailed product categories before the depreciation. In principle, this exposed rich consumers more to changes in the final consumer prices of foreign varieties. However, through an event study design, we show that the relative consumer price increase of foreign products compared to local alternatives was very muted.

⁴⁴ The subcategory level contains 14 different categories: Bakery/Cereal, Candy, Dairy, Dry Food, Fish, Fruit, Meat, Ready-made, Savory edibles, Seasonings, Vegetables, Coffee/Tea, Soft Drinks, and Waters.

⁴⁵ Judging from our preferred specification (which is column (3) of Table A.25), the elasticities range from -1.4 to -4.79 .

⁴⁶ The same result is also shown in Table A.26.

Whereas marginal costs of foreign products increased by 6–8% more relative to local varieties, this increase was offset by a decrease in the retail markups of foreign varieties by 3–4% relative to local varieties.

To analyze the change in the cost of living, we make assumptions about consumer preferences and decompose changes in the cost of living into (i) the price channel, (ii) the substitution channel, and (iii) the product variety channel. We estimate that the aggregate cost of living increased by 25% after four quarters. We also explore the distributional effects on the cost of living and show that the impact of the depreciation was less severe for rich consumers. In line with the event study, we show that consumers are more exposed to the relative cost shock because they allocate a larger share of their budget to foreign varieties, but they experience relatively lower retail markups after the shock. We estimate that rich consumers have lower elasticities of substitution which made them substitute away less intensively from varieties whose prices rose by more but it also means that they benefitted more from changes in product variety that occurred in the year following the depreciation.

Our results shed new light on the transmission mechanisms of exchange rate shocks to consumer prices and how offsetting movements in markups can dampen relative price adjustment after a depreciation. Furthermore, we show that heterogeneity in the price sensitivity of rich and poor consumers can translate into substantial cost-of-living inequality in the presence of large changes in product variety following international shocks.

DATA AVAILABILITY STATEMENT A part of the data that support the findings of this study are confidential and proprietary to Metro and AC Nielsen. Restrictions apply to the availability of these data, which were used under license for this study. Other data, including the programs used to execute the analysis, are openly available in repository openicpsr-202962 with the following DOI: <https://doi.org/10.3886/E202962V1>.

SUPPORTING INFORMATION

Additional supporting information may be found online in the Supporting Information section at the end of the article.

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